

Belpro

Beleženje Prostovoljstva
Digital Volunteer Diary for Slovenian NGOs

AI-Powered	WhatsApp-Native	GDPR-Compliant	Self-Hosted
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Architecture & Workflow Overview · 2025

What is Belpro?

Belpro is a smart, AI-powered digital assistant built specifically for Slovenian non-governmental organisations (NGOs). Its mission is simple: to make the legally required Volunteer Work Diary (Dnevnik prostovoljskega dela) as painless as possible for everyone involved.

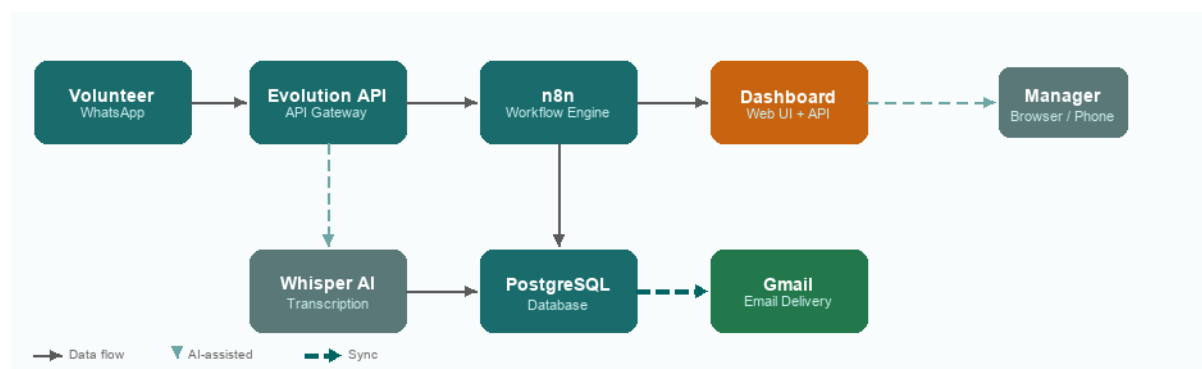
Under Slovenian law, volunteers receiving the Work Activity Allowance (Dodatek za delovno aktivnost) must log their activities monthly and submit proof to their local Centre for Social Work (CSD). Traditionally this meant paper forms, manual entries, and administrative overhead for the NGO. Belpro automates the entire process — from the moment a volunteer picks up their phone to the moment a signed PDF lands on the CSD desk.

Voice-first	Volunteers speak naturally in Slovenian — any dialect. AI does the rest.
WhatsApp	No new app to install. Uses the messaging platform volunteers already have.
Sovereign	Every NGO runs its own instance. No shared cloud, no third-party data access.
Auto-reports	Monthly PDF summaries generated automatically, ready for CSD submission.
Manager control	Every entry is reviewed and approved by the NGO manager before it is saved.

System Architecture

Belpro is built on a modern, self-hosted microservices stack, containerised with Docker and orchestrated locally. Every component is open-source, battle-tested, and self-hostable — no vendor lock-in, no subscription fees, no data leaving the NGO's own infrastructure.

The diagram below shows the high-level data flow: volunteers communicate via WhatsApp → Evolution API routes messages to n8n → n8n calls Whisper for transcription and writes approved records to PostgreSQL → the manager acts via the web dashboard or WhatsApp → Gmail delivers monthly reports.



Core Components

Component	Technology	Role
Messaging layer	WhatsApp / Evolution API	Real-time two-way communication with volunteers
AI Transcription	Faster-Whisper	Converts voice notes to text, on-device, CPU-optimised
Workflow engine	n8n	Orchestrates all business logic — no custom glue code
Message cache	Redis 7	Evolution API in-memory queue and cache (required dependency)
Database	PostgreSQL 18	Persistent, auditable storage of all volunteer records
Web dashboard	FastAPI + HTML/JS	Manager interface for approvals, analytics and reports
PDF engine	WeasyPrint	Generates print-ready monthly summaries
Email delivery	Gmail	Sends reports and notifications to volunteers

How It Works — The Volunteer Journey

From the volunteer's perspective, Belpro feels like a simple WhatsApp chat. Behind the scenes, a sophisticated AI pipeline captures, cleans, validates, and stores every entry with a complete audit trail.



① Record

The volunteer opens WhatsApp and sends a voice note describing what they did — in plain, everyday Slovenian. Any dialect is welcome. They can optionally attach a photo of completed work (a cleaned room, a tended garden bed). No forms. No apps. No training required.

② AI Cleans

Belpro's built-in speech recognition engine (Faster-Whisper) transcribes the voice note in real time, entirely on the NGO's own hardware. The transcript is then processed by the n8n workflow engine, which extracts the key facts — date, hours worked, activity type, location — and structures them into a clean record.

③ Confirm

Within seconds, the volunteer receives a neat summary back in WhatsApp: "Date: 12 May · 3 hours · Assisted with lunch at Dom starejših Trnovo." They can confirm with one tap, correct the entry if something was misheard, or cancel entirely. The volunteer is always in control.

④ Manager Approval

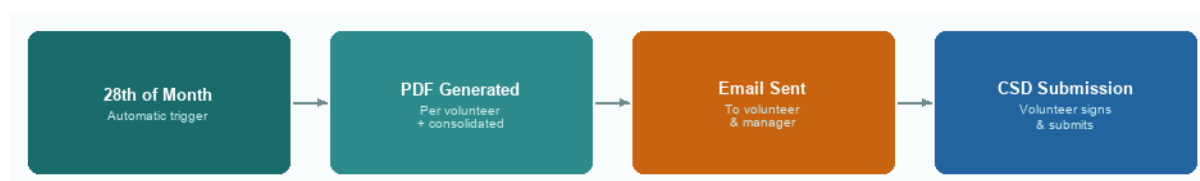
Once the volunteer confirms, the NGO manager receives an instant WhatsApp notification with the full entry details. A single tap approves or rejects. The manager can also view and action all pending entries via the web dashboard from any device.

⑤ Saved & Closed

An approved entry is written to the secure database with a full timestamp. Any attached photo is stored locally with its EXIF metadata (time and GPS location) intact — creating a tamper-evident audit trail. The volunteer receives a friendly confirmation message.

Monthly Reporting

At the end of each month, Belpro automatically generates all the paperwork the NGO needs — with zero manual effort.



Volunteer PDF

Each volunteer receives a personalised, print-ready PDF summarising their month's activities. The document is formatted to match the CSD submission requirements — the volunteer only needs to add their signature and hand it in.

Manager Report

The NGO manager receives a consolidated report covering all volunteers: total hours, a breakdown per person, and flagged entries. One document, complete picture.

Flexible Delivery

Reports can be emailed automatically (with the volunteer's consent), downloaded from the dashboard, or printed at the NGO office. Whatever works best for the volunteer.

Privacy & Compliance

GDPR & ZVOP-2	All data processing complies with Slovenian and EU data protection law.
Data sovereignty	All data stays on the NGO's own server. Nothing is sent to third parties.
Encryption	Sensitive personal data (EMŠO) is encrypted at rest using AES-256.
Photo privacy	Photos must show only work results — no faces or identifiable persons.
Retention	Data is kept only as long as required for CSD/IRSD inspections.
Audit trail	Every entry carries a complete, tamper-evident log of who approved what and when.